

Minimum-variance control for woofer–tweeter systems in adaptive optics

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The woofer–tweeter concept in adaptive optics consists in correcting for the turbulent wavefront disturbance with a combination of two deformable mirrors (DMs). The woofer corrects for temporally slow-evolving, spatially low-frequency, large-amplitude disturbances, whereas the tweeter is generally its complement, i.e., corrects for faster higher-order modes with lower amplitude. A special feature is that in general both are able to engender a common correction space. In this contribution a minimum-variance solution for the double stage woofer–tweeter concept in adaptive optics systems is addressed using a linear-quadratic-Gaussian approach. An analytical model is built upon previous developments on a single DM with temporal dynamics that accommodates a double-stage woofer–tweeter DM. Monte Carlo simulations are run for a system featuring an 8×8 actuator DM (considered infinitely fast), mounted on a steering tip/tilt platform (considered slow). Results show that it is essential to take into account temporal dynamics on the estimation step. Besides, unlike the other control strategies considered, the optimal solution is always stable. © 2010 Optical Society of America
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1. OPTIMAL CONTROL OF ASTRONOMICAL ADAPTIVE OPTICS SYSTEMS

Adaptive optics (AO) systems are now routinely used in ground-based telescopes to counter the blurring effects of atmospheric turbulence on image formation. They use a real-time controlled deformable mirror (DM) to generate a correction wavefront that is subtracted from the turbulent wavefront, using measurements of the residual phase provided by a wavefront sensor (WFS); for an overview of AO systems, refer to the review books [1,2].

In AO, the control objective is to minimize the variance of the residual wavefront after correction, since in doing so the well-known Strehl ratio is maximized [3].

A. Woofer–Tweeters

With the advent of the very large adaptive telescopes (using AO in the optical path) AO systems are meant to correct for unprecedentedly large amplitude disturbances. The latter increase considerably with D , the telescope diameter. To give a figure, stroke needs for Extremely Large Telescopes are more than ten-fold larger than those of present day AO systems. However, the mirror's stroke does not accompany the increase in diameter, and, additionally, correction is band-limited.

One possibility for achieving such levels of correction is to split the correction among two DMs—the commonly known woofer–tweeter (WT) [4], whose name originates from high-fidelity acoustic systems. With such a design, the woofer corrects for temporally slow-evolving, spatially low-frequency, large-amplitude disturbances, whereas the tweeter has a complementary behavior, i.e., faster high spatial and temporal frequencies with smaller amplitudes

(this is in accordance with the negative power law of the atmospheric power spectral density (PSD) of the phase disturbance). Combined, these two mirrors behave as a large-stroke, large-bandwidth mirror; see Fig. 1.

Several strategies address the control of the WT in AO: In [5,6] a modal static split of the modes between the woofer and the tweeter is proposed. The same is proposed in [7] in the spatial frequency domain, using a Fourier reconstructor that directly makes available those spatial frequencies. Both solutions assume infinitely fast mirrors. They consist mainly of splitting apart the spanned correction spaces so that they become disjoint. Though these solutions are attractive, no theoretical formalism has been set up to cast the problem as a minimum-variance (MV) control problem. Moreover, WT dynamics need to be properly taken into account.

Véran and Herriot treat the temporal regulation problem of the woofer–tweeter sharing the correction of the tip/tilt (TT) mode for the Thirty Meter Telescope–Narrow-Field Infrared Adaptive Optics System project. A sub-optimal controller is proposed based on a double integrator [8]. The woofer is a steering mirror with second-order dynamics and a cut-off frequency of about 20 Hz, and the tweeter is considered infinitely fast. A double integrator is proposed that tends to better attenuate the low frequencies in the residual. The parameters of the integrators are found by dichotomy with respect to stability constraints commonly accepted (in terms of gain and phase margins). However, since the woofer and the tweeter in this case span the same correction space, there is no explicit constraint that specifies the amount of correction assigned to each mirror. Another approach based on a model-reference adaptive system controller and a lattice filter

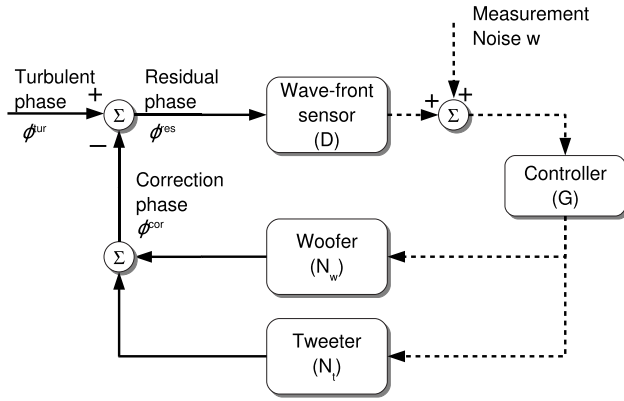


Fig. 1. Closed-loop control architecture. The correction phase is the joint contribution of the woofer's and tweeter's phases. Solid/dashed lines represent continuous/discrete-time variables.

controller is found in [9]. Again, no explicit minimization problem is solved.

Two main issues in controlling these devices are addressed in this contribution. One is related to taking into account the temporal dynamics of both mirrors. The second relates to the fact that mirrors may span overlapping spaces such that there is no uniqueness in control decisions to engender a given correction: Fig. 2.

The linear-quadratic-Gaussian (LQG) formalism [10] is used since the problem is linear, the AO cost-functional is described by a quadratic criterion, and disturbances are assumed normally distributed. The LQG formalism enables integration of the temporal dynamics of the phase by means of a linear and time-invariant (LTI) stochastic model and optimal (in the MV sense) estimation of the open-loop phase from closed-loop present and past measurements, optimality referring to the models that are chosen. The uniqueness of the solution is achieved by embedding the stroke constraints that translate into energy penalizations in the criterion. These turn out to be quite naturally incorporated by the LQG approach, since the minimization criterion relates to phase variance.

The solution presented here extends results of [11] referring to MV control of an AO system with DM dynamics to the double-corrector case.

Making use of a continuous stochastic model of the disturbance, a state-space model is constructed that yields a fully optimal MV solution in the form of a discrete-time LQG control. It further provides a means to (a) determine the optimal feedback controller and (b) assess the performance of any state-feedback linear discrete controllers. The control design is split into two steps: (i.) estimate the wavefront from noisy measurements and (ii.) compute the

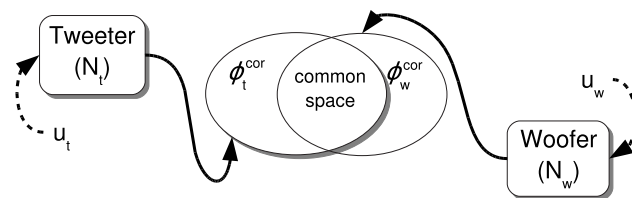


Fig. 2. Woofer and tweeter spanned spaces. Among the individual spatial correction spaces engendered by the woofer and the tweeter, a common space can be generated by both at the same time.

DM commands that optimally compensate for the estimated value of the turbulence. In a linear Gaussian framework, the *separation principle* guarantees global optimality of this two-step scheme [12,13].

The remainder of this paper is organized as follows. In Section 2 the instructive infinitely fast case is presented, later extended in Section 3 to the full-picture WT with non-negligible dynamics. Models of disturbance and DM for an illustrative example are discussed in Section 4. Numerical results are shown in Section 5 for a 8×8 actuator tweeter on a steering TT mount.

B. Main Notation

Throughout this paper, the value $\vartheta(t)$ of a continuous variable ϑ at sampling instants $t = kT_s$ will be denoted as $\vartheta_k \triangleq \vartheta(kT_s)$ and average values over the integration interval T_s by $\bar{\vartheta}_k \triangleq 1/T_s \int_{(k-1)T_s}^{kT_s} \vartheta(t) dt$. In the general case, x refers to a state vector with multiple entries. Matrices A_d , B_d , C_d , and D_d form the standard state-space system with state x_k and outputs z_k for the observations and z_k^{crit} to evaluate the criterion. State and measurement noises have covariance matrices, Σ_v and Σ_w , respectively. Measurements are concatenated in y and commands in u . Matrices D and N represent the WFS phase-to-measurements and the DM voltage-to-phase linear operations. Subscripts 'c' and 'd' are for continuous and discrete versions of the variables/operators, while tur, w, and t stand respectively for turbulent disturbance, woofer, and tweeter.

2. WOOFER-TWEETER OPTIMAL CONTROL: INFINITELY FAST CASE

Start from the MV phase residual criterion

$$\begin{aligned} \mathcal{J}^c(u) &\triangleq \lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^\tau \|\phi^{res}(t)\|^2 dt \\ &= \lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^\tau \|\phi^{tur}(t) - \phi^{cor}(t)\|^2 dt, \end{aligned} \quad (1)$$

where the residual phase ϕ^{res} is the difference between the turbulent and correction phases, respectively ϕ^{tur} and ϕ^{cor} (see Fig. 1). In Eq. (1) the L_2 norm is used over the telescope pupil Ω . This criterion is shown to maximize the well-known Strehl ratio [3].

When the DMs are infinitely fast, $\phi^{cor}(t) = Nu_k$ for all $t \in [kT_s, (k+1)T_s]$, where $N \in \mathbb{R}^{n_a \times n_\phi}$ is the DM influence matrix and T_s the loop sampling interval. n_a is the number of DM actuators, while n_ϕ is the number of phase modes. When N is full column rank, so that $N^T N$ is invertible, the criterion $\mathcal{J}^c$ is minimized by applying the optimal control decisions u_k in complete state information (CSI), i.e., assuming that $\bar{\phi}_{k+1}^{tur}$ is known [14],

$$u_k = (N^T N)^{-1} N^T \bar{\phi}_{k+1}^{tur}, \quad (2)$$

with

$$\bar{\phi}_{k+1}^{\text{tur}} = \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \phi^{\text{tur}}(t) dt. \quad (3)$$

In the general case of unknown turbulent phase disturbance, $\bar{\phi}_{k+1}^{\text{tur}}$ is to be estimated from measurements. The control theory's *separation principle* [15] guarantees optimality should the MV estimate of $\bar{\phi}_{k+1}^{\text{tur}}$ be used. In practice, it is obtained through a Kalman filter [16], assuming Gaussian distributions, a linear state-space model to obtain $\bar{\phi}_{k+1}^{\text{tur}}$, and *a priori* knowledge of $\Sigma_{\bar{\phi}}$ and Σ_w , the covariance matrices of $\bar{\phi}$ and measurement noise, respectively.

In the WT case, the control vector at time k is $u_k = \begin{pmatrix} u_k^w \\ u_k^t \end{pmatrix}$, so that the correction phase is $\forall t \in [kT_s : (k+1)T_s]$,

$$\phi^{\text{cor}}(t) = N_w u_k^w + N_t u_k^t = (N_w \quad N_t) \begin{pmatrix} u_k^w \\ u_k^t \end{pmatrix} = N u_k. \quad (4)$$

As the phase subspaces generated by the two mirrors generally overlap (Fig. 2), $N^T N$ is in the general case no longer invertible. This means that the original MV control problem has an infinity of equivalent solutions. Conversely, these additional degrees of freedom can be used to satisfy additional control objectives, leading to different possible control strategies.

One possible approach is to make maximum use of the woofer's capacity. This translates into minimizing J^c under the constraint that $\|u\|^2$ is minimal. The solution involves two successive orthogonal projections, leading to

$$u_k^w = (N_w^T N_w)^{-1} N_w^T \bar{\phi}_{k+1}^{\text{tur}}, \quad u_k^t = (N_t^T N_t)^{-1} N_t^T (\bar{\phi}_{k+1}^{\text{tur}} - N_w u_k^w). \quad (5)$$

Another solution is to modify the original MV problem by adding penalties on the control. This approach has been proposed in [17] to deal with stroke constraints. One may thus simply add to J^c a quadratic penalty term on the control u . The modified MV criterion thus becomes

$$J^c(u) \triangleq \lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^\tau \|\phi^{\text{res}}(t)\|^2 dt + \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} u_k^T \Delta R u_k, \quad (6)$$

with $\Delta R > 0$ in the form

$$\Delta R \triangleq \begin{pmatrix} \Delta R_w & 0 \\ 0 & \Delta R_t \end{pmatrix}. \quad (7)$$

In this case, the optimal solution becomes $u_k = (N^T N + \Delta R)^{-1} N^T \bar{\phi}_{k+1}^{\text{tur}}$. The ratio between ΔR_w and ΔR_t enables one to balance the control efforts between the two DMs. Also, when ΔR_w is small, the degradation in performance (compared with the penalty-free solutions) will be minimal. Furthermore, when $\Delta R_w = 0$ and $\Delta R_t \approx 0$, the resulting controller is nearly identical to the two-step projection of Eq. (5).

3. OPTIMAL WOOFER-TWEETER CONTROL WITH DEFORMABLE MIRROR DYNAMICS

The optimal control with DM dynamics stands upon two main results. The first is the existence of a discrete crite-

riterion equivalent to the continuous criterion on minimum residual variance, demonstrated in Subsection 3.A. The second is the construction of a discrete-time state-space model to represent the measurements and to obtain the output needed to evaluate the criterion and to establish the optimal state-feedback controller. That construction is presented in Subsection 3.B. Subsection 3.C discusses the optimal and two sub-optimal linear state-feedback solutions.

Without loss of generality, assume now that the woofer has temporal dynamics (ruled by a linear differential equation) and an infinitely fast tweeter ($\phi_t^{\text{cor}}(t) = N_t u^t(t)$).

Assume that the woofer correction phase is given by

$$\phi_w^{\text{cor}} = N_w p_w(t), \quad (8)$$

with $p_w(t)$ the woofer's instantaneous deformation on a basis of normalized tweeter influence functions. It is assumed that the tweeter shapes are all within $\text{Im}(N)$, the space spanned by the static influence functions.

The correction phase is the joint contribution of both DMs (Fig. 1), leading to

$$\phi^{\text{cor}}(t) = N_w p_w(t) + N_t u^t(t). \quad (9)$$

The woofer's instantaneous deformation is assumed to be the output of a continuous-time LTI state-space model in the form

$$\begin{cases} \dot{x}_w(t) = A_w x_w(t) + B_w u^w(t) \\ p_w(t) = C_w x_w(t) \end{cases}, \quad (10)$$

with the assumptions that (a) A_w is a stability matrix (all eigenvalues have strict negative real part), (b.) the pair (A_w, C_w) is observable, and (c) the DC gain between u^w and p_w is unitary. Note that assumptions (a)–(c) should essentially be regarded as the requirement that the woofer model be constructed according to the control theory's good practices.

Following the approach detailed in [11,18] for single mirrors and in [19] for WT, MV control using the LQG approach requires the determination of matrices Q_{wt} , S_{wt} , and R_{wt} to evaluate an equivalent discrete-time criterion in the form

$$\begin{aligned} J_{\text{dyn}}^d(u) &= \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} J_{\text{dyn}}^d(u)_k \\ &= \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} ((z_k^{\text{crit}})^T Q_{wt} z_k^{\text{crit}} + u_k^T R_{wt} u_k \\ &\quad + 2(z_k^{\text{crit}})^T S_{wt} u_k) \\ &= \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^M \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}^T \begin{pmatrix} Q_{wt} & S_{wt} \\ S_{wt}^T & R_{wt} \end{pmatrix} \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}, \quad (11) \end{aligned}$$

where the variable z_k^{crit} is shown to be formed by three components:

$$z_k^{\text{crit}} \triangleq \begin{pmatrix} \bar{\phi}_{k+1}^{\text{tur}} \\ \bar{\phi}_{k+1}^{\text{tur}} \\ x_k^w \end{pmatrix} \triangleq \begin{pmatrix} \frac{1}{T_s} \int_0^{T_s} \phi^{\text{tur}}(kT_s + s) ds \\ \frac{1}{T_s} \int_0^{T_s} e^{sA_w^T} C_w^T N_w^T \phi^{\text{tur}}(kT_s + s) ds \\ x^w(kT_s) \end{pmatrix}. \quad (12)$$

The discrete-time quadratic criterion J_{dyn}^d will satisfy the following properties:

1. $Q_{\text{wt}} - S_{\text{wt}} R_{\text{wt}}^{-1} S_{\text{wt}}^T \geq 0$, with $R_{\text{wt}} = R_{\text{wt}}^T > 0$.
2. $\arg \min_{\mathcal{U}} J^c(u) = \arg \min_{\mathcal{U}} J_{\text{dyn}}^d(u)$.

The first property, under suitable assumptions on the turbulent phase, guarantees the existence of a unique (and easily implementable) LQG discrete-time controller minimising J_{dyn}^d . The second property, on the other hand, shows that this discrete LQG controller is optimal with respect to the original continuous-time criterion J^c .

A. Equivalent Discrete-Time Criterion

The construction of the equivalent discrete-time criterion follows the approach used in [11] for a single DM with dynamics. Its starting point is the expression of the average residual phase variance between two successive sampling times. Using the expression of the correction phase in Eq. (9), one gets

$$\frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{res}}(t)\|^2 dt = \frac{(9)}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t) - N_w p_w(t) - N_t u^t(t)\|^2 dt, \quad (13)$$

which is shown in Appendix A.

On the other hand, for a constant input $u^w(t) = u_k^w, \forall t \in [kT_s, (k+1)T_s]$, the exact solution of $p_w(t)$ is given by

$$p_w(t) = C_w e^{(t-kT_s)A_w} x_k^w + (I + C_w e^{(t-kT_s)A_w} A_w^{-1} B_w) u_k^w. \quad (14)$$

The next step is to substitute this expression of $p_w(t)$ in Eq. (13) and to evaluate the resulting integrals. This enables expression of the average residual variance in Eq. (13) as a deterministic function of u_k and z_k^{crit} . However, this function is not a proper quadratic form. To recover the positiveness of the corresponding discrete-time criterion J_{dyn}^d , it is therefore necessary to add appropriate terms independent of the control u . Once these somewhat tedious manipulations have been performed (see details in Appendix A), the resulting LQ weighting matrices become

$$Q_{\text{wt}} \triangleq \begin{pmatrix} \lambda(2N_w N_w^T + N_t N_t^T) & 0 & 0 \\ 0 & \lambda(I + 2B_w^T A_w^{-T} A_w^{-1} B_w) & -I \\ 0 & -I & Q_0 \end{pmatrix} \geq 0, \quad (15)$$

$$S_{\text{wt}} \triangleq \begin{pmatrix} -N_w & -N_t \\ -A_w^{-1} B_w & 0 \\ S_0 & T_{\text{wt}} N_w^T N_t \end{pmatrix}, \quad (16)$$

and

$$R_{\text{wt}} \triangleq \begin{pmatrix} R_0 + \Delta R_w & (I + G_{\text{wt}}) N_w^T N_t \\ N_t^T N_w (I + G_{\text{wt}})^T & N_t^T N_t + \Delta R_t \end{pmatrix} > 0, \quad (17)$$

where

$$T_{\text{wt}} = \frac{1}{T_s} \int_0^{T_s} (C_w e^{tA_w})^T dt, \quad (18)$$

$$G_{\text{wt}} = \frac{1}{T_s} \int_0^{T_s} (C_w e^{sA_w} A_w^{-1} B_w)^T dt, \quad (19)$$

and

$$R_0 \triangleq \frac{1}{T_s} \int_0^{T_s} (I + C_w e^{sA_w} A_w^{-1} B_w)^T N_w^T N_w (I + C_w e^{sA_w} A_w^{-1} B_w) ds, \quad (20)$$

$$S_0 \triangleq \frac{1}{T_s} \int_0^{T_s} e^{sA_w^T} C_w^T N_w^T N_w (I + C_w e^{sA_w} A_w^{-1} B_w) ds, \quad (21)$$

$$Q_0 \triangleq \frac{1}{T_s} \int_0^{T_s} e^{sA_w^T} C_w^T N_w^T N_w C_w e^{sA_w} ds. \quad (22)$$

The definition of Q_{wt} includes an arbitrary positive constant λ . It is shown (see details in Appendix A) that the following properties hold:

- For all $\lambda \geq 0$, $\arg \min_{\mathcal{U}} J^c(u) = \arg \min_{\mathcal{U}} J_{\text{dyn}}^d(u)$.
- For large enough values of λ , $Q_{\text{wt}} - S_{\text{wt}} R_{\text{wt}}^{-1} S_{\text{wt}}^T \geq 0$, with $R_{\text{wt}} = R_{\text{wt}}^T > 0$.
- For $\lambda = 0$, $J^c(u) = J_{\text{tur}} + J_{\text{dyn}}^d(u)$.

Also note that the infinitely fast DM case is recovered as a limit case of the results above. To prove it, consider a set of DM models in the form of Eq. (10) depending on a parameter θ such that for $\theta \rightarrow 0$, the maximum value of the real part of all eigenvalues of A_w tends to $-\infty$, so that the dynamics, loosely speaking, become “infinitely fast.” Routine matrix calculations show that the corresponding asymptotic behavior for the weighting matrices is

$$\lim_{\theta \rightarrow 0} Q_0 = 0, \quad \lim_{\theta \rightarrow 0} S_0 = 0, \quad \lim_{\theta \rightarrow 0} R_0 = N_w^T N_w. \quad (23)$$

Likewise, in the absence of additional quadratic penalty on u , in other words for $\Delta R = 0$,

$$\lim_{\theta \rightarrow 0} R_{\text{wt}} = \begin{pmatrix} N_w^T N_w & N_w^T N_t \\ N_t^T N_w & N_t^T N_t \end{pmatrix}. \quad (24)$$

This matrix is not strictly positive-definite except in the special case where $N_t^T N_w = 0$, which means that the correction spaces spanned by the woofer and the tweeter are orthogonal. Conversely, when the two correction spaces share a non-empty intersection, the uniqueness of the optimal control is no longer guaranteed when $\Delta R = 0$.

In practice, a suitable choice for the additional quadratic penalization is

$$\Delta R = \begin{pmatrix} \epsilon_t \sigma_\phi^{-2} N_w^\top N_w & 0 \\ 0 & \epsilon_w \sigma_\phi^{-2} N_t^\top N_t \end{pmatrix}, \quad (25)$$

where the ratio $\epsilon_t/\epsilon_w \approx \|u_w\|^2/\|u_t\|^2$ defines the control effort applied to each mirror. The penalties are chosen as a function of the total disturbance variance the devices are supposed to correct for, projected onto the controls space through $N_w^\top N_w$ and $N_t^\top N_t$. This choice of ΔR will be used in the application presented in Section 4.

B. Control-Oriented Model

Consider the standard discrete-time state-space model of the form

$$\begin{cases} x_{k+1} = A_d x_k + B_d u_k + \Gamma_d v_k \\ z_k = C_d x_k + w_k \\ z_k^{\text{crit}} = C_d^{\text{crit}} x_k \end{cases}, \quad (26)$$

where the first line represents the state evolution equation, the second the noisy delayed measurements provided by a wavefront sensor, and the last the extraction of the coordinates of x_k needed to evaluate the discrete-time criterion J^d in Eq. (11).

The disturbance is modeled as the output of a stochastic LTI system whose input is a Gaussian white noise with PSD Σ_v ,

$$\begin{cases} \dot{x}^{\text{tur}}(t) = A_{\text{tur}} x^{\text{tur}}(t) + v(t) \\ \phi_{\text{tur}}(t) = C_{\text{tur}} x^{\text{tur}}(t) \end{cases}. \quad (27)$$

The coefficients of A_{tur} are chosen from the short-term correlation function of the disturbance modes modeled [11].

Focusing on the specific arrangement of measurements and controls in AO—Fig. 3—the measurement at $t = kT_s$ is given by

$$y_k = D \frac{1}{T_s} \int_{(k-2)T_s}^{(k-1)T_s} (\phi^{\text{tur}}(t) - N_w p_w(t) - N_t u_{k-2}^t) dt + w'_k, \quad (28)$$

$$\begin{aligned} &= D \bar{\phi}_{k-1}^{\text{tur}} - D N_t u_{k-2}^t - D N_w \left[\frac{C_w}{T_s} (e^{T_s A_w} - I) A_w^{-1} x_{k-2}^w \right. \\ &\quad \left. + \left(\frac{C_w}{T_s} (e^{T_s A_w} - I) A_w^{-2} B_w + I \right) u_{k-2}^w \right] + w'_k, \end{aligned} \quad (29)$$

where w'_k is a zero-mean additive Gaussian white noise.

Equation (13) is obtained by replacing $p_w(t)$ by its expression in Eq. (14) and further simplifying. The measurement equation in the space-state model of Eq. (26) used a modified version of Eq. (13) such that

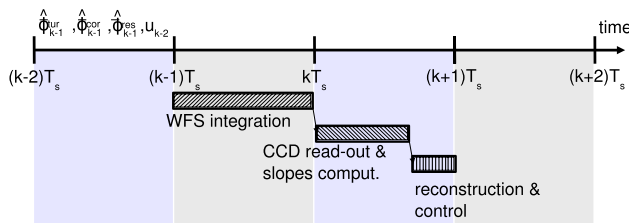


Fig. 3. (Color online) Chronogram of operations.

$$z_k = y_{k+1}. \quad (30)$$

For the reasons pointed out in [11], this allows for a more compact state representation, being mathematically equivalent to using y_k directly if the delay is assigned to the control instead of the measurement.

Using the exact discretization of the disturbance model of Eq. (27), the DM model of Eq. (10), the pseudo-measurement of Eq. (30), and the performance output of Eq. (12), the state-space model of the standard form of Eq. (26) is fully determined by

$$x_k \triangleq \begin{pmatrix} x_{k+1}^{\text{tur}} \\ \bar{\phi}_{k+1}^{\text{tur}} \\ \bar{\phi}_{k+1}^{\text{crit}} \\ x_k^w \\ u_{k-1}^t \\ \bar{\phi}_k^{\text{tur}} \\ x_{k-1}^w \\ u_{k-1}^w \end{pmatrix}, \quad A_{\text{wt}} \triangleq \begin{pmatrix} e^{T_s A_{\text{tur}}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \Xi & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \Theta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{T_s A_w} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (31)$$

where the fifth component of the state is the tweeter's commands.

$$B_{\text{wt}} \triangleq \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ A_w^{-1} (e^{T_s A_w} - I) B_w & 0 \\ 0 & I \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ I & 0 \end{pmatrix}, \quad \Gamma_{\text{wt}} \triangleq \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (32)$$

$$\begin{aligned} C_{\text{wt}} \triangleq D \left(0 \quad 0 \quad 0 \quad 0 \quad -N_t \quad I \quad -\frac{N_w C_w}{T_s} (e^{T_s A_w} - I) A_w^{-1} \right. \\ \left. - N_w \left(\frac{C_w}{T_s} (e^{T_s A_w} - I) A_w^{-2} B_w + I \right) \right), \end{aligned} \quad (33)$$

where $D \in \mathbb{R}^{n_m \times n_\phi}$ is the phase-to-measurements matrix.

The performance output is obtained from the state x_k through $C_{\text{wt}}^{\text{crit}}$, with

$$C_{\text{wt}}^{\text{crit}} \triangleq \begin{pmatrix} 0 & I & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (34)$$

Matrices Ξ and Θ used to obtain, respectively, $\bar{\phi}_{k+1}^{\text{tur}}$ and $\bar{\phi}_{k+1}^{\text{crit}}$ are derived in [11], this model being an extension of the one derived therein.

Using this model, the off-line computational burden amounts to computing two Riccati equations: an estimation and a control Riccati equation. Though this may seem uninteresting and discourages AO designers from using LQG approaches, the estimation Riccati equation is

computed on the reduced set of stochastic states, i.e., the first three entries of the state vector in Eq. (31). The same applies to the control Riccati equation, solved for the first four entries (i.e., in full-state information, since the separation theorem applies). In a purely modal control scenario, the former amounts to solving Riccati's equations of dimensions $n_\phi + n_{ord,\phi} n_\phi + n_{ord,w} n_w$ for the estimation and $n_\phi + n_{ord,\phi} n_\phi + 2n_{ord,w} n_w$ for the control, where n_ϕ is the number of atmospheric modes, $n_{ord,\phi}$ is the order of the atmospheric dynamical model, $n_{ord,w}$ is the order of the woofer model, and n_w the number of woofer-controlled modes. If $n_\phi = n_w$ and second-order models are considered, then the sizes of the matrices are $5n_\phi$ and $7n_\phi$.

In terms of real-time application, under the same assumptions, accounting for the particular multiplication of the innovation plus the update and prediction steps presents a computation burden of $O(\alpha n_\phi^2)$ with factor $\alpha = 21$. Thus, compared to the oversimplified integrator controller, the optimal one demands about ten-fold more computational power.

C. Optimal and Sub-Optimal Solutions

The optimal LQ controller is of the form

$$u_k = -\mathcal{K}_\infty \hat{x}_{k|k-1}, \quad (35)$$

with $\mathcal{K}_\infty$ defined as

$$\mathcal{K}_\infty = (R_w + \mathcal{B}_d^\top P_\infty \mathcal{B}_d)^{-1} (\mathcal{B}_d^\top P_\infty \mathcal{A}_d + S_w), \quad (36)$$

determined from the (unique) solution to the discrete-time control algebraic Riccati equation (ARE):

$$P_\infty = Q_w + \mathcal{A}_d^\top P_\infty \mathcal{A}_d - (\mathcal{A}_d^\top P_\infty \mathcal{A}_d + S_w) \times (R_w + \mathcal{B}_d^\top P_\infty \mathcal{B}_d)^{-1} (\mathcal{B}_d^\top P_\infty \mathcal{A}_d + S_w). \quad (37)$$

The MV estimate $\hat{x}_{k|k-1}$ is computed recursively as the output of a Kalman filter adjusted to the state-space model. The criterion $J^c(u)$ in Eq. (1) can be analytically evaluated using the results of Appendix B. Two controllers are proposed as follows:

- A first sub-optimal solution consists in using the LQG solution that neglects the mirrors' dynamics. This sub-optimal solution in this new context of application with non-negligible dynamics has been found optimal in Section 2 for infinitely fast mirrors. For such cases, a simple orthogonal projection of the phase onto the mirror space is performed,

$$\begin{pmatrix} u_k^w \\ u_k^t \end{pmatrix} = (N^\top N + \Delta R)^{-1} N^\top \bar{\phi}_{k+1}^{\text{tur}} = P \bar{\phi}_{k+1}^{\text{tur}}, \quad (38)$$

where $N \triangleq (N_w^\top, N_t^\top)$ and ΔR is defined in Eq. (25).

- An alternative solution, called intermediate from this point forward, is to apply to the woofer a command

$$u_k^w = (N_w^\top N_w)^{-1} N_w^\top \bar{\phi}_{k+1}^{\text{tur}} = P_w \bar{\phi}_{k+1}^{\text{tur}} \quad (39)$$

and to the tweeter the remainder after correction by the woofer,

$$u_k^t = P_t \left[\bar{\phi}_{k+1}^{\text{tur}} - \left(N_w u_k^w - \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} N_w p_w(t) dt \right) \right]. \quad (40)$$

At this stage, two possibilities are found: either take into account or not the dynamical behavior of the woofer on the estimation (done by a Kalman filter).

1. Take into account the dynamics in the estimation of the state. Then both commands considered together are close to those of an orthogonal projection on two infinitely fast mirrors. Such strategy is indeed rather interesting since the controller synthesis becomes considerably simpler: there is no requirement to explicitly estimate x_k^{tur} and $\bar{\phi}_{k+1}^{\text{tur}}$ in the state of Eq. (31). This can be seen by noting that when replacing $p_w(t)$ in Eq. (40) by its expression in Eq. (14), just $\bar{\phi}_{k+1}^{\text{tur}}$ and x_k^w need be known. A closely related control strategy presented in [5] proposes to apply to the tweeter at each instant the remainder of the disturbance left uncorrected by the woofer. This implies driving the tweeter at a higher frame rate during the interval of duration T_s . Since the tweeter has the greatest density of actuators, the outcome is a considerable increase in the real-time computational requirements. For this reason, this approach is not treated here.

2. Neglect the dynamics only in the estimation of the state but not in the computation of the control. The first simulations with this strategy turn out to be an under-performing, which emphasizes the need to use enhanced estimators for the phase.

Given the arguments that precede, in the remainder of this paper, three control strategies are compared against each other, i.e., the optimal state-feedback controller with gain $\mathcal{K}_\infty$ is compared with an intermediate sub-optimal controller that only partially takes dynamics into account (in the state estimation), sharing the orthogonal projection of the estimate onto the DM with a sub-optimal solution that completely neglects the mirror's dynamics. This sub-optimal solution in this new context with non-negligible dynamics has been found optimal in Section 2 for infinitely fast mirrors. For such cases, a simple orthogonal projection of the phase onto the mirror space is performed.

4. SIMULATION OF A DEFORMABLE MIRROR ON A TIP/TILT PLATFORM

In what follows, the simulation conditions chosen are those of a telescope with a primary mirror diameter of 30 m.

Inspired by the NFIRAOS AO system for the Thirty-Meter Telescope (TMT), a DM mounted on a steering platform with temporal dynamics is addressed. A simpler version is modeled with an 8×8 actuator tweeter. Addressing the joint contribution of steering mirrors and DM is all the more important since this specific design is commonly used in AO. Hitherto, given the small size of the steering mirrors, they have been considered infinitely fast.

Tip/tilt (TT) is the simplest of all aberrations. Yet it is often the most difficult to correct for. This is because its amplitude is often much larger than that of any other ab-

errations, representing roughly 90% of the total uncorrected wavefront variance [20]. In addition, the effect of wind buffeting on the telescope's structure gives rise to large amplitude disturbances that need to be completely corrected for. For example, preliminary studies for the European ELT (42 m diameter) reveal disturbances 2 to 3 orders of magnitude larger than atmospheric TT. For the TMT (30 m diameter) wind buffeting is approximately of the same amplitude as atmospheric TT. Such difference is due to the mechanical structure of the telescope's enclosure (commonly called the dome), with the TMT's being smaller and more protective.

For the purposes of the illustration, without any loss of generality, it is assumed that TT is in the common space engendered by the WT (in Fig. 1). The TT is considered the mean slope of the phase over the pupil (in two directions in quadrature). This mode is also commonly called angle of arrival. Additionally to the large amplitude, the interest of controlling this particular mode is that, when using a Hartmann–Shack wavefront sensor, it can easily be computed from the measurements by just taking their mean.

Assume now that a fully featured AO system is being addressed, where several other modes are in the common space, and moreover, the total number of modes to control (whether with or without temporal dynamics) are to be dealt with. Two strategies to control one such a system not restrained to the TT can be envisioned with the material proposed here:

- (a) Spatially separate the TT from the higher-order modes in a VLT-SPHERE-like manner by doing orthogonal projections onto the TT and its complementary;
- (b) Treat the modes in an all-at-once manner.

In case (a) the spatially spanned spaces of the woofer and tweeter are disjoint. Therefore the tweeter will not generate parasite TT that could degrade the performance and eventually lead to the saturation of the actuators (by assigning corrections of opposite signs to the WT).

In case (b) there is no spatial distinction of the modes, i.e., both the woofer and the tweeter are allowed to generate corrections in an overlapping correction space. A caveat is that the computational requirements required if many modes are addressed tend to increase notably. Mechanical finite-element-method models indicate that temporal dynamics concentrate on a reduced modal set; therefore, since we are addressing only a relatively small number of modes that present a parasite temporal dynamics, the computational requirements issue does not play a crucial role.

A. Disturbance Linear Model

Figure 4 depicts the PSD of the atmospheric TT with a total 21.4 milliarcseconds (mas) rms, obtained from the linear combination of spectra in directions parallel and perpendicular to the wind velocity proposed in [21]. Since wind-buffeting for the TMT is of the same order of magnitude, this example is representative of the overall disturbance.

A second-order model has been defined by fitting the first points of the auto-correlation of the TT in the vicinity of zero. A broader discussion on the choice of the distur-

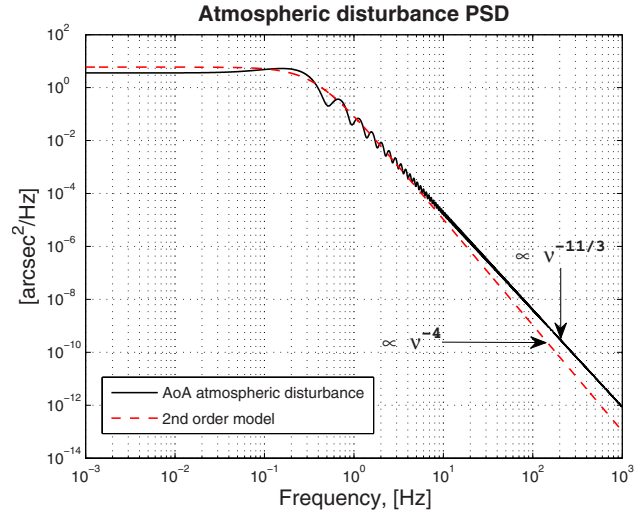


Fig. 4. (Color online) PSD of the disturbance with a total of 21.4 mas rms. Second-order model (normalized amplitude) used based on fitting the auto-correlation of the disturbance at scales in the vicinity of zero. $D/r_0=420@0.5\ \mu\text{m}$, $L_0\approx 37\text{ m}$, $V_i=[13.2;8.6;7.1]\text{ m/s}$, $\theta_i=[0,45,90]\text{ deg}$, for a vertical turbulence profile with relative strengths $[0.67\ 0.22\ 0.11]$.

bance model parameters can be found in [11]. The model obtained (dashed line in Fig. 4) closely follows the disturbance PSD in both low- and high-frequency regimes.

B. Woofer Model

A second-order model is chosen for the steering mirror since it grasps the nature of the mechanical dynamics, which is well approximated by a second-order differential equation,

$$\ddot{p}_w(t) + 2\xi\omega_n\dot{p}_w(t) + \omega_n^2 p_w(t) = \omega_n^2 u^w(t), \quad (41)$$

where ξ is the damping coefficient and $\omega_n=2\pi f_n$ is the natural resonance pulsation (in rad/s), obtained from the resonant frequency f_n in hertz. In terms of state-space representation, Eq. (41) converts to

$$\begin{aligned} \mathcal{A}_w &= \begin{pmatrix} 0 & 1 \\ -\omega_n^2 & -2\xi\omega_n \end{pmatrix}, \quad \mathcal{B}_w = \begin{pmatrix} 0 \\ \omega_n^2 \end{pmatrix} \\ \mathcal{C}_w &= (1 \ 0), \quad \mathcal{D}_w = 0, \end{aligned} \quad (42)$$

with $\xi=0.35$ and $\omega_n=88\text{ rad/s}$ chosen as default values. The bandwidth at -3 dB is $\approx 20\text{ Hz}$. The block diagram of the model used for the numerical illustration is depicted in Fig. 5. The influence matrix is taken as $N_w=I$; that is, the continuous deformation directly gives the correction phase, $\phi_w^{\text{cor}}(t)=p_w(t)$, $\forall t \geq 0$.

C. Tweeter Model

The tweeter is an 8×8 actuator DM with bi-cubic influence functions aligned in a Cartesian grid (Fig. 6) with 20% cross-coupling.

The separable bi-cubic influence functions $\text{IF}_c(x,y) = \text{IF}_c(x)\text{IF}_c(y)$ are given by

$$IF_c(x) = \begin{cases} 1 + (4c - 2.5)|x|^2 + (-3c + 1.5)|x|^3 & \text{if } |x| \leq 1 \\ (2c - 0.5)(2 - |x|)^2 + (-c + 0.5)(2 - |x|)^3 & \text{if } 1 < |x| < 2 \\ 0 & \text{otherwise} \end{cases}, \quad (43)$$

where the constant c defines the cross-coupling between neighboring actuators. Figure 7 shows a radial cut for $c = 0.2$.

The matrix N_t is a concatenation of the projection of the influence function onto the tilt mode. The i th column is written

$$N_t(:, i) = \frac{1}{S_\Omega} \int_{\Omega} TT(x, y) IF_i(x, y) dx dy, \quad (44)$$

where S_Ω is the total surface of the pupil, $TT(x, y)$ is the tilt mode, and $IF_i(x, y)$ is the i th influence function re-orthonormalized, i.e.,

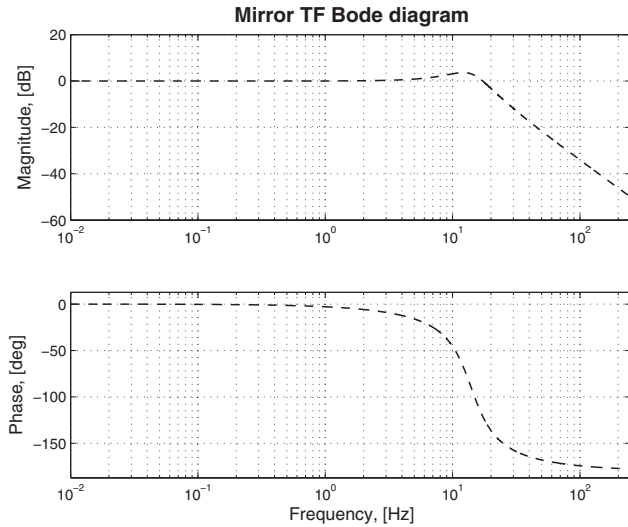


Fig. 5. (Color online) Bode diagram of the DM transfer function (TF). A resonance of a factor 1.525 is observable at the frequency of $f_r = 13.1$ Hz. Model parameters are $\xi = 0.35$ and $\omega_n = 88$ rad/s.

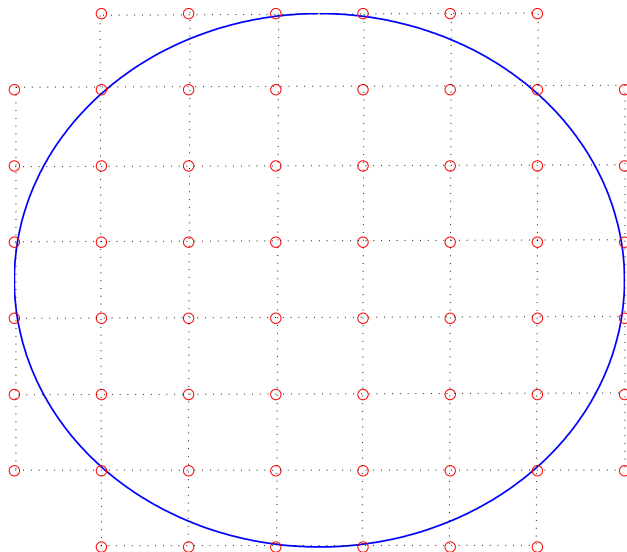


Fig. 6. (Color online) Actuator location. Circles, actuators; solid line, telescope aperture.

$$\begin{cases} \frac{1}{S_\Omega} \int_{\Omega} IF_j(x, y) IF_i(x, y) dx dy = 1 & \text{if } i = j; \\ 0 & \text{otherwise} \end{cases}. \quad (45)$$

5. SAMPLE NUMERICAL RESULTS

To illustrate the principle of the MV optimal control, the common space in Fig. 1 was chosen to correspond to the TT. Disturbance and mirror models are presented in Section 4.

The DM commands are computed from the TT modal commands by back-projecting the solution onto the DM influence functions using the pseudo-inverse of N_t . For the case of a single tip or tilt mode, i.e., with N_t is a row-vector, the pseudo-inverse is straightforwardly obtained as

$$N_t^\dagger = \frac{N_t^T}{\|N_t\|^2}. \quad (46)$$

Using bi-cubic influence functions allows one to represent TT modes exactly. When considering the telescope's pupil and the actuators' locations, though, TT modes are no longer perfectly fitted. The fitting errors project onto higher-order modes (orthogonal to TT) and thus are not accounted for in the total error budget. See Section 4 for a discussion on how to deal with this higher-order error.

Using results developed in Appendix B, Fig. 8 compares the performance in milliarcseconds (mas) rms of the optimal solution with the intermediate [Eqs. (39), (40)] and the sub-optimal one (recall that it consists in using an

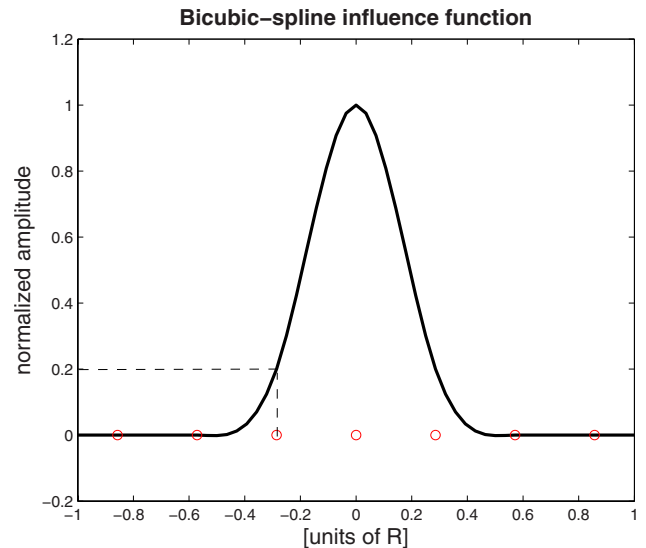


Fig. 7. (Color online) Bi-cubic-spline influence function with 20% cross-coupling. Circles are the actuator locations, normalized by the telescope radius R .

LQG solution that neglects the dynamics). The impact of the noise at the edges of the pupil is addressed for five distinct frame rates. Results are computed for the two TT modes (assumed statistically independent [20]).

The optimal controller ensures the highest performance in terms of lowest residual phase and residual noise variances and is stable by construction. The sub-optimal controller is seen to have limited stability and to present poorer levels of performance. Note that though close, optimal and sub-optimal performance are to be analyzed in light of the diffraction limit of the telescope. Observing at $1\ \mu\text{m}$, it means 6.88 mas and 4.9 mas for a 30 m and 42 m telescope primary diameter, respectively. The gain of 1 mas rms thus means 15% and 20% of the diffraction, a non-negligible improvement from which scientific instruments are to benefit accordingly. All solutions were defined so as to assign woofer commands that were 100-fold more energetic than those of the tweeter. In turn, the DM corrects for 2.5 mas rms, which is equivalent to $\approx 0.09\ \mu\text{m}$ rms actuator displacement at the edges of the pupil.

An intriguing question concerns the overall performance achieved by the conjunction of two mirrors. When varying the repartition of controls from half/half to all-to-the-woofer, the performance remains practically unchanged in the optimal case, a result shown in Fig. 9. The leftmost points correspond to a tweeter required to have a $\sim 0.027\ \mu\text{m}$ rms displacement at the pupil edge, a rather small value. This means that in the optimal case the TT platform can theoretically deal with the whole TT disturbance, in which case the DM would be left alone to correct for higher-order disturbances.

These figures change when considering the sub-optimal solutions. The presence of a tweeter is indeed advantageous since the woofer dynamics (only partially or not at all taken into account on the controller) can thus be overcome by the infinitely fast tweeter. A variation of the or-

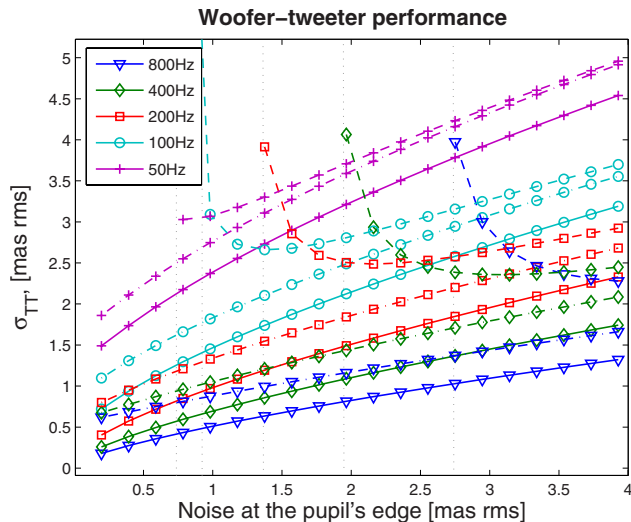


Fig. 8. (Color online) Performance of the optimal and the sub-optimal controller as a function of the measurement noise. Five frame rates are used. Solid lines are for the optimal, dot-dashed lines are for the intermediate, and dashed lines are for the sub-optimal controller. The ratio of penalties is $\epsilon_t/\epsilon_w=100$. Vertical dotted lines indicate the stability limits of the sub-optimal solution.

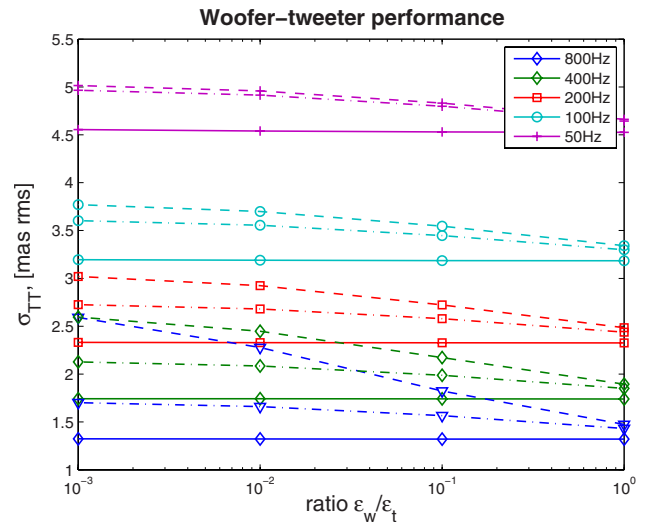


Fig. 9. (Color online) Performance as a function of the ratio of control energy assigned to the woofer and to the tweeter. Noise at the telescope's edge is 200 nm rms, which is in terms of TT ≈ 3.9 mas rms. On the abscissa, the penalties are taken as $\epsilon_w=1$ and ϵ_t variable. Solid lines are for the optimal, dot-dashed lines are for the intermediate, and dashed lines are for the sub-optimal controller.

der of 0.5–1 mas rms is observed. Note also that when the tweeter is assigned the whole correction i.e., the woofer is shortcut, the performance converges to that of the optimal solution. In other words, the optimal solution is that of a system with infinitely fast mirrors only.

An important issue to underline is that no physical limitation on the stroke is actually applied, the control amplitudes being fed to the WT as computed from the controller (penalized, though). Taking those limitations into account leads to a constrained minimization problem. A related problem with DM actuator saturation is found in [17].

A. Temporal Trajectories

Using the Monte Carlo simulator described in [11], the temporal trajectories of both the woofer and the tweeter can be determined. Figure 10 compares the effects of taking and not taking DM dynamics into account. The optimal controller gives rise to smoother responses with the woofer being seemingly delayed with respect to the disturbance. This happens because the 20 Hz bandwidth makes it quite slow. Nonetheless, the presence of the tweeter allows for the total correction phase to closely follow the input disturbance.

The intermediate controller is able to avoid the woofer's strong oscillations but makes poorer use of the tweeter. It is interesting to notice, though, that the tweeter's response is in this case smoother than with the optimal controller, which can be seen as an advantage. However, the price to pay is a loss in performance, 0.34 mas rms (see Fig. 9) worse than the one obtained by the optimal controller.

On the other hand, not taking the dynamics into account results in a particularly strong oscillatory behavior that tends to degrade the performance, this being representative of the results obtained in the previous section.

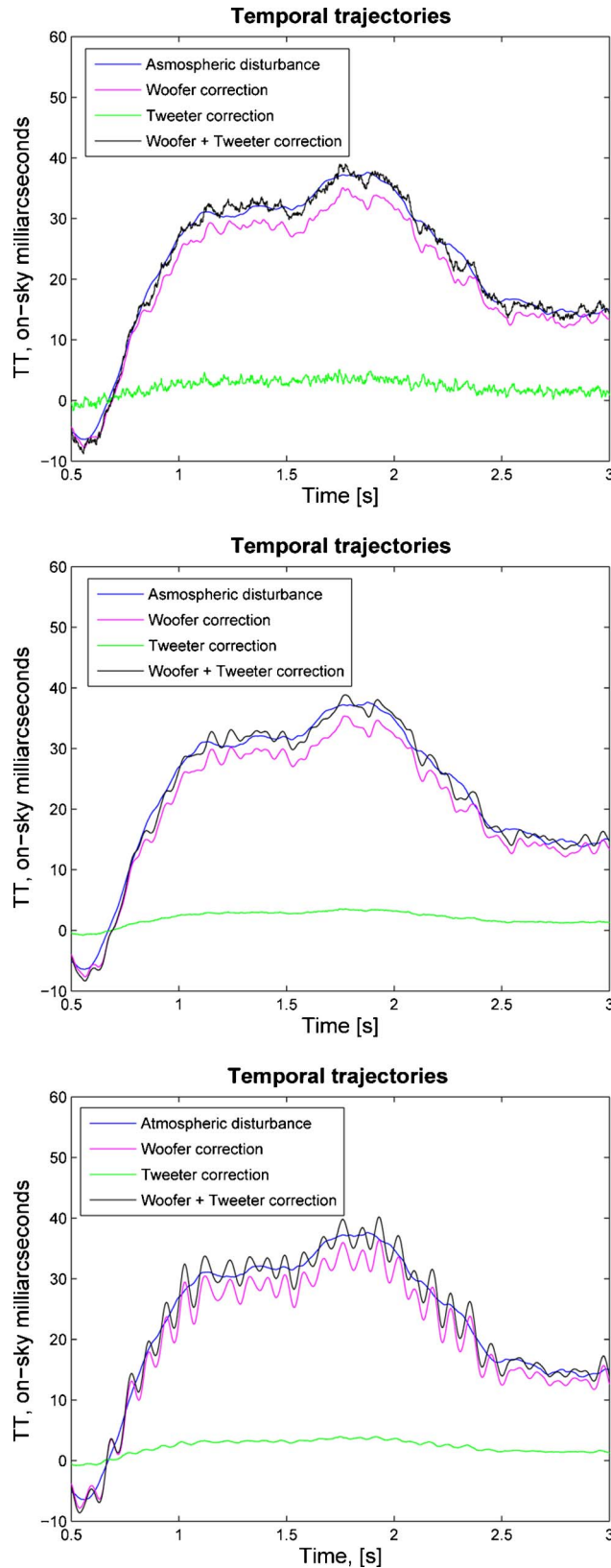


Fig. 10. (Color online) Temporal trajectories for the WT for the optimal (top), intermediate (center), and sub-optimal (bottom) solutions, with $T_s = 1/400$ s and $\epsilon_t/\epsilon_w = 100$. From 30 s of simulation, the empirical values were found (in mas rms): Top: $\phi^{res} = 1.742$, $u_w = 21.46$, $u_t = 2.42$; Centre: $\phi^{res} = 2.085$, $u_w = 21.67$, $u_t = 2.166$; Bottom: $\phi^{res} = 2.448$, $u_w = 21.69$, $u_t = 2.167$.

6. SUMMARY

The MV control problem for the WT concept has been established in the general case where the two mirrors span correction spaces with non-empty intersection. The optimal solution is a reconstructed state feedback, namely a linear quadratic Gaussian controller, where the optimal control and estimation gains are computed using algebraic Riccati equations, the state estimate being obtained as the output of a Kalman filter, similarly to the method in [11]. This formulation allows one to analytically evaluate a system's ultimate performance on the basis of the theoretical calculation of the variance for a wide range of configurations featuring linear dynamics of disturbance and DM. Performance of sub-optimal WT controllers can also be evaluated using the same tool. Many combinations of temporal and spatial bandwidths may therefore be explored for the DMs' function, for example, of a specific choice of sampling frequencies.

Simulations for the TT case of a DM on a steering mount (a NFIRAOS-type system) highlight the advantages of using an optimal approach that takes into account the woofer's dynamics. Physical stroke constraints are translated into extra penalization on the control energy, so that the optimal solution distributes the control effort in a user-defined ratio. The effect of hard-actuator saturation has not been evaluated. This could be done using the tools proposed here, which provide a means of simulating the temporal behavior of a particular system, including various nonlinear features such as control clipping that cannot be accounted for in the analytical performance evaluation.

The analysis of the numerical results provided herein shows that using only the woofer is enough if the temporal dynamics are known and optimally taken into account. Moreover, the optimal solution is always stable and provides better optical performance for all the combinations of parameters that have been tested (sampling frequency and measurement noise).

Two sub-optimal controllers have been presented: orthogonal projection of a predicted disturbance that does not account for DM's dynamics (dubbed "sub-optimal controller") and an intermediate solution that consists in orthogonally projecting onto the DMs the predicted disturbance (dubbed "intermediate controller"), obtained from a Kalman filter that includes the DM's dynamics. Of course, if one is willing to increase the tweeter rate, other sub-optimal controllers could be envisioned, thus allowing one to compensate the woofer dynamics at a higher frame rate with simple filters.

Results obtained with the two sub-optimal control strategies presented in this paper, chosen among other tested strategies, suggest that accounting for the DM's dynamics in the estimation process is essential to obtain stable controllers, irrespective of the working point, i.e., resonant frequency, dumping factor, and signal and noise variances. This is the starting point of the intermediate solution, which has the advantage of avoiding the determination of the model describing the interaction between the disturbance and the DM's response (i.e., $\bar{\varphi}^{tur}$). However, it enhances the optical performance of the sub-optimal strategy, with still worse handling of control effort than with the optimal control.

APPENDIX A: SOLUTION DERIVATION

Start from Eq. (13) by noting that

$$\begin{aligned}
 & \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t) - N_w p_w(t) - N_t u^t(t)\|^2 dt \\
 &= \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} (\phi^{\text{tur}}(t) - N_w p_w(t) - N_t u^t(t))^T (\phi^{\text{tur}}(t) \\
 &\quad - N_w p_w(t) - N_t u^t(t)) dt \\
 &= \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \phi^{\text{tur}}(t)^T \phi^{\text{tur}}(t) + p_w(t)^T N_w^T N_w p_w(t) \\
 &\quad - 2\phi^{\text{tur}}(t)^T N_w p_w(t) - 2\phi^{\text{tur}}(t)^T N_t u^t + 2p_w(t)^T N_t^T N_t u^t \\
 &\quad + (u^t)^T N_t^T N_t u^t dt. \quad (\text{A1})
 \end{aligned}$$

Denote as F the transfer function from u to p , i.e., $F(s) = C_w(sI - A_w)^{-1}B_w + D_w$. From point (c.) in Eq. (10), $F(0) = D_w - C_w A_w^{-1} B_w = I$. Using this and the identity $\int_0^s e^{vA_w} dv = A_w^{-1}(e^{sA_w} - I)$, one gets

$$p_w(t) = C_w e^{(t-kT_s)A_w} x_k^w + (I + C_w e^{(t-kT_s)A_w} A_w^{-1} B_w) u_k^w, \quad (\text{A2})$$

which is Eq. (14). Then

$$\begin{aligned}
 & p_w(kT_s + s)^T N_w^T N_w p_w(kT_s + s) \\
 &= [C_w e^{sA_w} x_k^w + (I + C_w e^{sA_w} A_w^{-1} B_w) u_k^w]^T N_w^T N_w \\
 &\quad \times [C_w e^{sA_w} x_k^w + (I + C_w e^{sA_w} A_w^{-1} B_w) u_k^w] \\
 &= (x_k^w)^T (e^{sA_w} C_w^T N_w^T N_w C_w e^{sA_w}) x_k^w + 2(x_k^w)^T \\
 &\quad \times [e^{sA_w} C_w^T N_w^T N_w (I + C_w e^{sA_w} A_w^{-1} B_w)] u_k^w + (u_k^w)^T \\
 &\quad \times [(I + C_w e^{sA_w} A_w^{-1} B_w)^T N_w^T N_w (I + C_w e^{sA_w} A_w^{-1} B_w)] u_k^w. \quad (\text{A3})
 \end{aligned}$$

By integrating over $t \in [kT_s, (k+1)T_s]$, one thus gets

$$\begin{aligned}
 & \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} p_w(t)^T N_w^T N_w p_w(t) dt \\
 &= (x_k^w)^T Q_0 x_k^w + 2(x_k^w)^T S_0 u_k^w + (u_k^w)^T R_0 u_k^w, \quad (\text{A4})
 \end{aligned}$$

with Q_0 , S_0 , and R_0 defined by Eq. (22).

Proceeding likewise for the remainder terms in Eq. (A1),

$$\begin{aligned}
 & \phi^{\text{tur}}(kT_s + s)^T N_w p_w(kT_s + s) \\
 &= \phi^{\text{tur}}(kT_s + s)^T N_w [C_w e^{sA_w} x_k^w + (I + C_w e^{sA_w} A_w^{-1} B_w) u_k^w] \\
 &= \phi^{\text{tur}}(kT_s + s)^T N_w C_w e^{sA_w} (x_k^w + A_w^{-1} B_w u_k^w) \\
 &\quad + \phi^{\text{tur}}(kT_s + s)^T N_w u_k^w. \quad (\text{A5})
 \end{aligned}$$

Integrating over $t \in [kT_s, (k+1)T_s]$,

$$\begin{aligned}
 & \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} p_w(t)^T N_w^T \phi^{\text{tur}}(t) dt \\
 &= (u_k^w)^T N_w^T \bar{\phi}_{k+1}^{\text{tur}} + (x_k^w + A_w^{-1} B_w u_k^w)^T \bar{\varphi}_{k+1}^{\text{tur}}, \quad (\text{A6})
 \end{aligned}$$

where

$$\bar{\phi}_{k+1}^{\text{tur}} = \frac{1}{T_s} \int_0^{T_s} \phi^{\text{tur}}(kT_s + s) ds \quad (\text{A7})$$

is the average phase over a sampling interval T_s , and

$$\bar{\varphi}_{k+1}^{\text{tur}} = \frac{1}{T_s} \int_0^{T_s} e^{sA_w} C_w^T N_w^T \phi^{\text{tur}}(kT_s + s) ds \quad (\text{A8})$$

is the result of the cross-product obtained in developing the quadratic criterion.

Continuing the integration of the individual terms of Eq. (A1),

$$-\frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \phi^{\text{tur}}(t)^T N_t u_k^t dt = -(\bar{\phi}_{k+1}^{\text{tur}})^T N_t u_k^t, \quad (\text{A9})$$

and finally

$$\begin{aligned}
 & \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} [C_w e^{(t-kT_s)A_w} x_k^w + (I + C_w e^{(t-kT_s)A_w} A_w^{-1} B_w) u_k^w]^T \\
 &\quad \times N_w^T N_t u_k^t dt \\
 &= (x_k^w)^T T_{wt} u_k^t + (u_k^w)^T (N_w^T N_t + G_{wt}) u_k^t, \quad (\text{A10})
 \end{aligned}$$

where T_{wt} and G_{wt} are defined by Eqs. (18) and (19). The last term is written

$$\frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} (u_k^t)^T N_t^T N_t u_k^t dt = (u_k^t)^T N_t^T N_t u_k^t. \quad (\text{A11})$$

This establishes that

$$J^c(u)_k = \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t)\|^2 dt + \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}^T \begin{pmatrix} Q_{wt} & S_{wt} \\ S_{wt}^T & R_{wt} \end{pmatrix} \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix} dt, \quad (\text{A12})$$

where

$$\begin{pmatrix} Q_{wt} & S_{wt} \\ S_{wt}^T & R_{wt} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -N_w & -N_t \\ 0 & 0 & -I & -A_w^{-1} B_w & 0 \\ 0 & -I & Q_0 & S_0 & T_{wt} N_w^T N_w \\ -N_w^T & -B_w^T A_w^{-T} & S_0^T & R_0 + \Delta R_w & (I + G_{wt}) N_w^T N_t \\ -N_t^T & 0 & N_w^T N_w T_{wt}^T & N_t^T N_w (I + G_{wt})^T & N_t^T N_t + \Delta R_t \end{pmatrix}. \quad (\text{A13})$$

This formula corresponds to the weighting matrices in

Subsection 3.A for $\lambda=0$. The completion of the quadratic

form is performed as in [11], noting that for some $\lambda > 0$,

$$\begin{pmatrix} Q_0 & S_0 & T_{wt} N_w^T N_w \\ S_0^T & R_0 + \Delta R_w & (I + G_w) N_w^T N_t \\ N_w^T N_w T_{wt}^T & N_t^T N_w (I + G_w)^T & N_t^T N_t + \Delta R_t \end{pmatrix} \geq \frac{1}{\lambda} I > 0. \quad (A14)$$

Using this property, appropriate Schur complements can be added to Q_{wt} , ensuring that $Q_{wt} - S_{wt} R_{wt}^{-1} S_{wt}^T \geq 0$, with $R_{wt} = R_{wt}^T > 0$.

APPENDIX B: CRITERION ANALYTICAL EVALUATION

As shown in Appendix A, for $\lambda = 0$ the identity $J^c(u) = J_{\text{tur}} + J_{\text{dyn}}^d(u)$ holds. Since $z_k^{\text{crit}} = C_d^{\text{crit}} x_k$ and $u_k = -K_{\infty} \hat{x}_{k|k-1}$:

$$\begin{aligned} J_{\text{dyn}}^d(u)_k &= \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}^T \begin{pmatrix} Q_{wt} & S_{wt} \\ S_{wt}^T & R_{wt} \end{pmatrix} \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix} \\ &= \begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix}^T \begin{pmatrix} C_d^{\text{crit}T} Q_{wt} C_d^{\text{crit}} & -C_d^{\text{crit}T} S_{wt} K_{\infty} \\ -K_{\infty}^T S_{wt}^T C_d^{\text{crit}} & K_{\infty}^T R_{wt} K_{\infty} \end{pmatrix} \begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix}. \end{aligned}$$

Furthermore, because ϕ^{tur} is an ergodic process, the average disturbance energy over an infinite-horizon is almost surely (a.s.) equal to its expected value [22], i.e.,

$$\lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^{\tau} \|\phi^{\text{tur}}(t)\|^2 dt = E(\|\phi^{\text{tur}}(t)\|^2), \quad (B1)$$

at any time t , including sampling times $t = kT_s, k \in \mathbb{N}$. $E(\cdot)$ denotes mathematical expectation. Thus,

$$\lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^{\tau} \|\phi^{\text{tur}}(t)\|^2 dt = \text{trace}(C_{\text{tur}}' C_{\text{tur}}'^T \Sigma_x), \quad (B2)$$

where $C_{\text{tur}}' \triangleq (C_{\text{tur}} \ 0 \ \cdots \ 0)$. Likewise, the infinite-horizon criterion $J_{\text{dyn}}^d(u)$ is almost surely equal to the expected value of $J_{\text{dyn}}^d(u)_k$. This leads to:

$$\begin{aligned} J^c(u) &= E(\|\phi^{\text{tur}}(t)\|^2) + E((z_k^{\text{crit}})^T Q_{wt} z_k^{\text{crit}} + u_k^T R_{wt} u_k \\ &\quad - 2(z_k^{\text{crit}})^T S_{wt} u_k). \end{aligned} \quad (B3)$$

As a result, the continuous-time criterion $J^c(u)$ can be written almost surely as

$$\begin{aligned} J^c(u) &= E \left(\begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix}^T \begin{pmatrix} C_{\text{tur}}' C_{\text{tur}}'^T + C_d^{\text{crit}T} Q_{wt} C_d^{\text{crit}} & -C_d^{\text{crit}T} S_{wt} K_{\infty} \\ -K_{\infty}^T S_{wt}^T C_d^{\text{crit}} & K_{\infty}^T R_{wt} K_{\infty} \end{pmatrix} \begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix} \right) \\ &\quad \times \begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix}^T = \text{trace}(W \Sigma_{\xi}), \end{aligned} \quad (B4)$$

where Σ_{ξ} is the steady-state covariance of the state-vector $\begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix}$.

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